SIGNA MONITOR CALIBRATION / GAMMA ADJUSTMENT

EFFECTIVITY:

All Signa Horizon LX Operator workspace monitors with model numbers M2400FR, MDW321 / MDW321 Plus and LCD2000 / LCD2010. All cameras which comply with the M952 interface standard.

PURPOSE:

The operator workspace monitor must be properly adjusted for the filmed image to accurately represent the console monitor displayed image.

RELATED FMI'S:

None.

FURNISHED MATERIALS:

FMI Instructions.

MANPOWER REQUIREMENTS:

One Service Engineer for one and half (1.5) hours, no travel.

Camera Vendor for one (1) hour for contrast / density adjustments.

SPECIAL TOOLS & TEST EQUIPMENT:

None.

FIELD SUPPLIED MATERIALS:

None.

DISPOSAL OF SOFTWARE OR MANUALS:

None.

1- OVERVIEW

The operator workspace monitor must be properly adjusted for the filmed image to accurately represent the console monitor displayed image. This procedure will describe how to adjust the monitor to match the camera. **Once the monitor is re-calibrated, it is essential to re-calibrate the camera before the system is used for filming.**

1-2 FMI Schedule Considerations

Although the steps outlined below are possible for a qualified GE Service Engineer to perform for both Camera and Display, it is recommended that the procedure is performed with the support of the Camera Vendor field engineer. Unless the site requires the introduction of this FMI earlier, the FMI should be scheduled to coincide with the next normally scheduled Camera PM. The GE Service Engineer should notify the Customer of the need for this procedure to be performed to coincide with the next PM schedule. The monitor adjustments will take no loner than 30 minutes to be performed. Typically, 15 minutes is all that is needed. Camera adjustments should be performed immediately after the monitor adjustments and before clinical use. The time required for adjusting the camera varies with each product manufacturer.

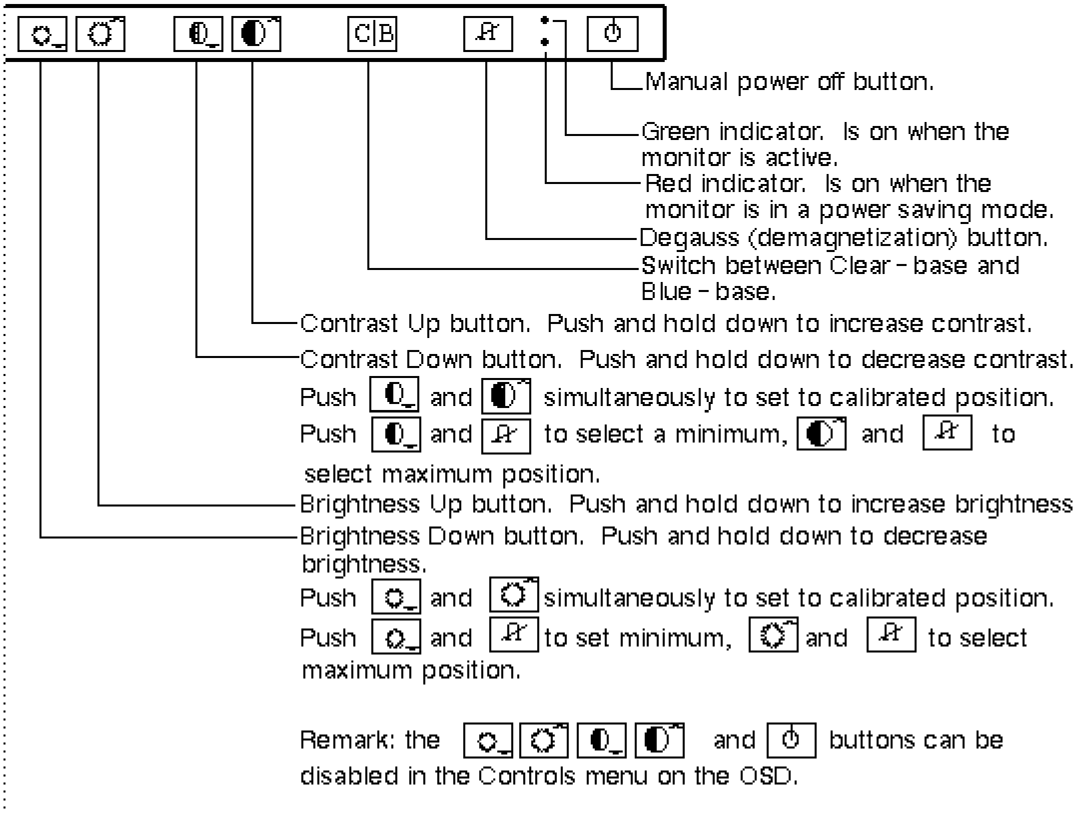
2- MONITOR CALIBRATION CONTROLS

2-1 Model MDW 321 Controls (including MDW321 Plus)

The controls for the Model MDW 321 are shown in Illustration L1.

Notes

For the Model MDW 321, the user controls are disabled. For this model, if power is cycled the contrast and brightness changes will be permanently saved and will not revert back to the factory default values. These controls will need to be enabled for this procedure and then disabled. For the MDW 321, follow the procedure for enabling and disabling the user controls.



MDW 321 COLOR MONITOR CALIBRATION CONTROLS

ILLUSTRATION L1

2-2 Enabling/Disabling User Controls for the MDW 321

1. Bring up the OSD (On Screen Display) by selecting the up contrast and down brightness buttons simultaneously.
2. Use the up contrast or down contrast to scroll through the menu to select the entry for Controls.
3. Push the down brightness button to enter the Controls submenu.
4. Upon entering the controls submenu, the User Controls entry should be highlighted, (if not use the up/down contrast button to select User Controls).
5. To toggle the User Controls to on/off, select the C/B button.
6. Once User control is set, you may back out of the OSD menus by repeatedly selecting the up brightness control. Alternately, after 30 seconds with no activity, the menu will disappear.

2-3 Model LCD2000/LCD2010 Controls

The controls for Models LCD2000 and LCD2010 are accessible from the On-Screen Manager™. To access the On-Screen Manager OSM™ , press any of the control buttons (▲,▼,**+**,**-**) to get to the main menu. The function of the OSM™ controls on the front of the monitor is described in Table T-1. Use the Front panel control functions to assess and modify the brightness and contrast settings per this instruction. Once the calibration is complete, use the OSM Lock Out to fix the settings.

Notes

Allow the Model LCD2000/LCD2010 monitor to warm up for 20 minutes before performing any adjustments.

2-4 Enabling/Disabling Controls for the LCD2000/LCD2010

This control completely locks out access to all OSM control functions. When attempting to activate controls while in Lock Out mode, a message screen will appear indicating that the controls are locked out.

- To activate the Lock Out function, press PROCEED, and ▼ simultaneously.
- To de-activate the Lock Out function, press PROCEED, and ▲ simultaneously.

TABLE T-1
FRONT PANEL CONTROL FUNCTIONS - MODEL LCD2000

Control	Main Menu	Sub-Menu
EXIT	Exits the OSM™ controls.	Exits to the OSM™ controls main menu.
CONTROL ▲/▼	Used to access main menu. Moves the highlighted area up/down to select one of the controls.	Moves the highlighted area up/down to select one of the controls.
CONTROL +/-	Has no function	Moves the bar in the + or **-** direction to increase or decrease the adjustment.
PROCEED	Proceeds to the selected menu choice (indicated by the highlighted area).	Activates Auto Adjust feature. In Display Mode, opens additional window.
RESET: The currently highlighted control to the factory setting.	Resets all the controls within the highlighted menu.	Resets the highlighted control.

Note

When **RESET** is pressed, a warning window will appear allowing you to cancel the reset function.

Adjustment information is **not** lost when the power is removed from the monitor.

3- VERIFY AMBIENT LIGHTING CONDITIONS

In the review area and operator workspace area, verify that the ambient lighting conditions are adjusted to a minimum level. In the operator workspace area, there should be only sufficient light for safely operating the system.

In the review area and operator workspace area, verify that light-boxes are not emitting light, or are properly masked, when not displaying film. This will be a source of excessive glare. In both the review area and the operator workspace area, verify that there is no source of glare for reviewing films or setting up the images for film. For example, windows should not allow direct light (blinds should be closed).

Note that both the operator workspace area and the review area artificial lighting type should be equivalent.

4 MONITOR CALIBRATION

4-1 Verify and set the color temperature

A large variation in color temperature between the monitor and the review station light-box/film can cause the operator to read the film differently. Set the color temperature to 9300K.

4-2 Modifying the GAMMA Value

The GAMMA value is modified to optimize the contrast level of the image mid-tones to more closely represent the same contrast that is filmed. The result of incorporating this gamma correction value will make the film image and the image displayed on the host monitor similar in appearance.

1. Make sure the system has power.

   a. Power up the operator workspace, which includes the monitor, LCD panel, and both the host computer and PC.

   b. After IRIX has booted (applications may or may not be running), log in to the host computer. See Procedure for Bringing the Host Computer Up and Down for details on how to boot the host computer.

   c. After the host computer is at applications level, initiate a Winterm window from the GE puzzle-piece icon.

   d. Log in as root. Edit the colorPix.cfg file, if color monitor, by entering, “jot /usr/g/colorPix/colorPix.cfg” or edit the grayPix.cfg file, if B/W monitor, by entering, “jot /usr/g/grayPix/grayPix.cfg”.

   e. Modify the numerical value for gamma to 1.1, for color CRT monitors, 1.05 for LCD monitors, and 0.9 for the B/W monitor. Save the changes. Exit the jot file editor. This modifies the gamma value such that if the system reboots, the value is not lost.

   f. At the Winterm command prompt, enter “gamma 1.1” for color CRT monitor, “gamma 1.05” for color LCD monitor, “gamma 0.9” for B/W CRT monitor. This is in addition to “step e”. This step immediately changes the gamma value without the need to reboot.

You are now ready to proceed with setting contrast and brightness on the display as outlined in the next section.

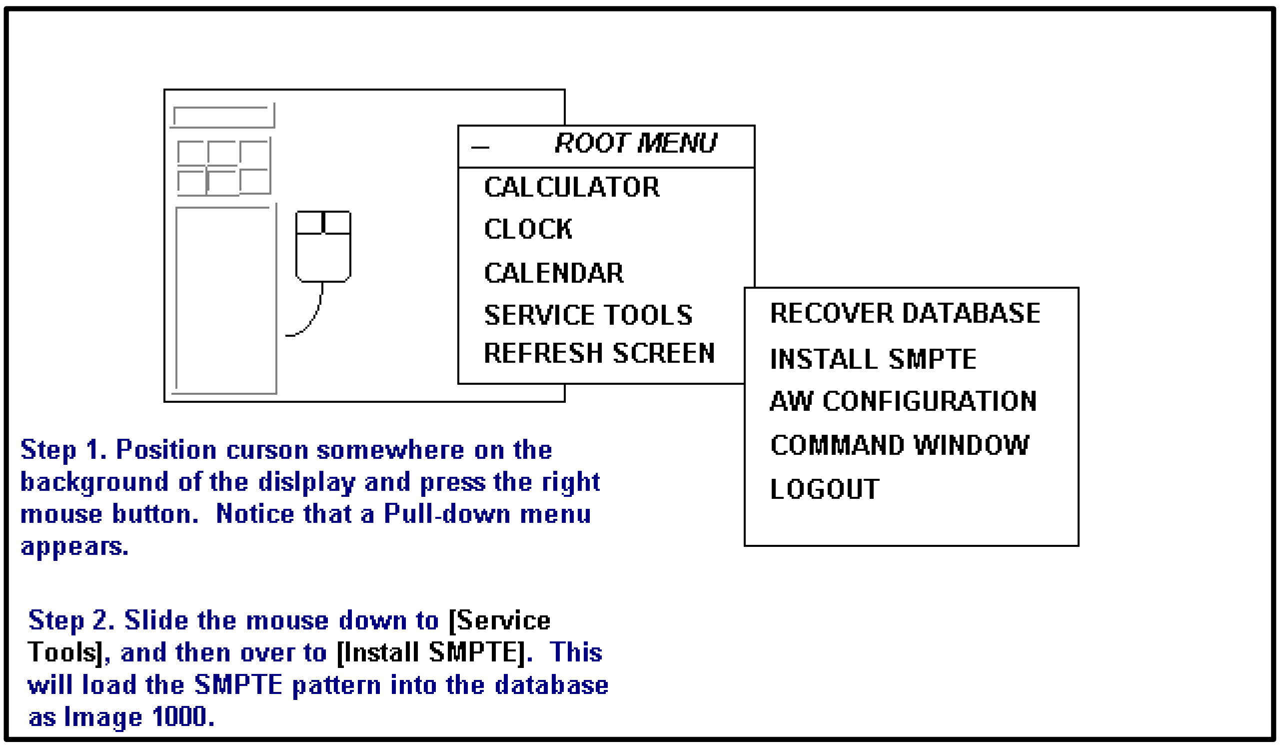
4-3 CONTRAST AND BRIGHTNESS

4-3-1 Displaying the SMPTE Pattern

The SMPTE (Society of Motion Picture and Television Engineers) test pattern is used to provide a standard image for calibrating the display.

This test pattern is available on the Operator Workstation Host Computer after has fully booted.

1. Once the host computer is at applications level and the gamma is set to 1.1 for a color CRT, 1.05 for a LCD, 0.9 for a B/W monitor, install and display the SMPTE pattern. See Illustration L8 for installing and Illustration L9 for displaying the SMPTE pattern. See Illustration L2482A for sample test pattern.

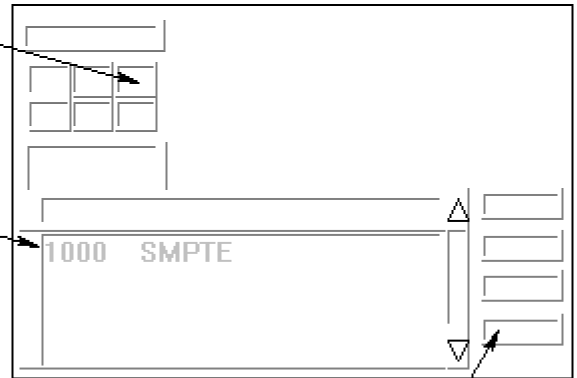


INSTALLING THE SMPTE PATTERN
ILLUSTRATION L8

Step 1. Point to and click on the Display (AW) icon. Notice that the Browser comes up.

Step 2. After the Browser comes up, use the scroll bar on the right side of the display to find Image 1000 SMPTE.

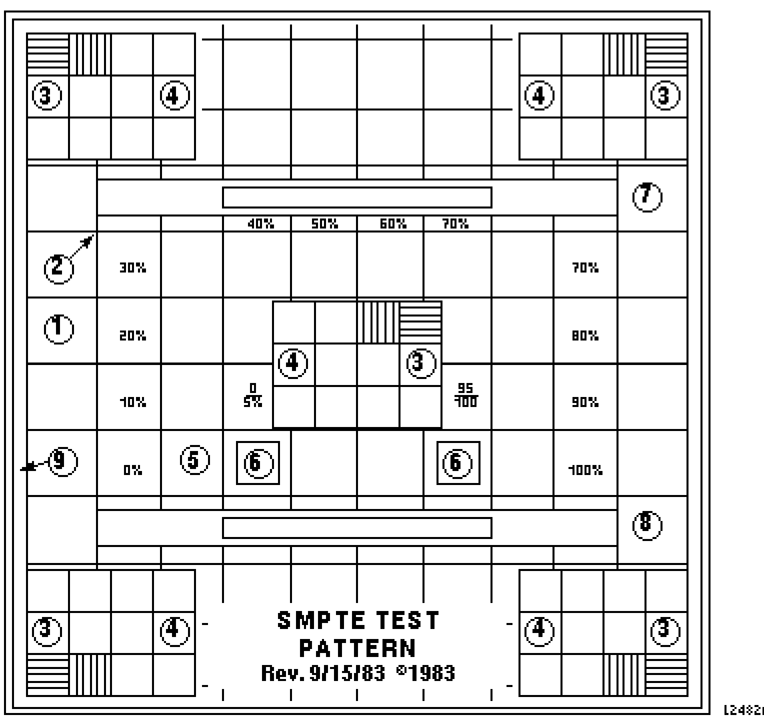
Step 3. Point to and single-click on the SMPTE entry.



Step 4. Point to and single-click on the Full Viewer to display the SMPTE so that it is full screen size.

Step 5. To get back from displaying the SMPTE pattern, hit <ESC>.

DISPLAYING THE SMPTE PATTERN
ILLUSTRATION L9



SAMPLE SMPTE TEST PATTERN
ILLUSTRATION L2

2. With the cursor inside the displayed image, type "**ww 100**" to set window width to 100.

3. With the cursor inside the displayed image, type "**w1 1024**" to set window level to 1024.

5- CONTRAST AND BRIGHTNESS ADJUSTMENT

1. With the SMPTE Pattern Displayed, adjust the brightness control of the display monitor to the extreme minimum brightness, see Illustration L2.

2. Next, adjust the contrast to the maximum setting that does not cause visible tearing or smearing of the pattern or alpha-numeric characters. Note that with the LCD panel, you should also adjust the contrast to the maximum without any visible washout of the pattern or alpha-numeric characters.

3. Next, adjust the brightness control of the display monitor until the scanning raster for the 0% is barely visible and that the 5% and 95% patches are visible.(item 6). Note that you may need to re-adjust the contrast if tearing or smearing of the pattern or alpha-numeric characters occurred (items 1, 2, 5, 6, 7,& 8). As with the contrast adjustment, note that with the LCD panel you should also adjust the brightness to the maximum without any visible washout of the pattern or alpha-numeric characters.

6- CAMERA CALIBRATION

This procedure describes the steps necessary to verify and set the Camera parameters. **Once the display is re-calibrated, it is essential to re-calibrate the camera before the system is used for filming.** Although the steps outlined below are possible for a qualified GE Service Engineer, it is recommended that the following procedure be performed with the Camera Vendor field engineer.

6-1 DASM interpolaton method

From the service desktop, select the install icon to modify the DASM. Set the DASM interpolation method to linear.

6-2 Optical Density Matching

Note that for optimal reviewing, the light-box luminance of the diagnostic region of the film should be in the range of 50 to 500 nits, as viewed through the light-box.

To calibrate the light-box and film to meet this value, the camera setting for maximum optical density should be varied. A good starting position is a maximum density of 2.8. This is a good starting target for camera calibration. However, the final OD (Operator Display) settings may be refined by the radiologists performing the image review.

6-3 Contrast Optimization

1. With the maximum/minimum optical densities set for the review area's light-box, select a look-up table for your camera that will produce a perceivably linear gray scale for the associated light-box and ambient light conditions. Note that the DICOM 3.14 Standard specifies the Barten's curve for linear perception. It is recommended that the manufacturer base perceptual linearity on this curve.

2. Film the SMPTE pattern on a 1-on-15 format display. Verify that the 5% and the 95% levels are visually equivalent. If not, perform a Contrast test with the SMPTE pattern. Select the new contrast setting from the contrast image set. A good value for the Imation Dryview is 3. Use the Camera's calibration procedure to set the contrast setting. Ensure that the camera maintains a perceivably linear gray scale.

Note

Filming the SMPTE pattern for contrast calibration may be optional for the camera manufacturer.

3. Ask the technologist to display a clinical image and set window and level controls for desired appearance. A sagittal or axial head image is a good image to start out with.
4. Capture the image on the keypad or host control interface.
5. Print a Contrast Test film. Ask the technologist to select the image that most closely represents the displayed image on the monitor.
6. Observe the image number below the selected image and set the Contrast control to this value.

7- ANATOMICAL FILMING

This portion of the procedure requires the technologist to verify the camera settings with true anatomical images.

Film representative anatomical images to confirm the settings. The image set should include T1 and T2 head images, joint images and c-spines. Observe the accuracy of the low-tones, mid-tones and high-tones. If a filmed image is found not to be not equivalent, re-calibrate the camera based on the customer's evaluation.

8 - FINAL STEPS

8-1 Site Turnover

1. Do one Head and one Body scan to insure proper system operation.
2. Record FMI installation on the "FMI Installation Record" located inside the pocket attached to the inside of the PDU front door.
3. Close service dispatch to close FMI file.